

Chapter 8 — Conclusion

8.1 The Research Question Restated

This dissertation asked: across European populations in the period 2007–2024, what is the shape of the education–fertility gradient by country, and what conditions predict its steepness, its shape, and the individual-level mechanism through which it operates? Four hypotheses operationalised this question. H1 predicted that the relationship between education and period eTFR is negative; H2 predicted that the relationship is non-monotonic, with fertility ceasing to decline or rising at the upper end of the educational distribution; H3 predicted that cross-country variation in the gradient is better predicted by cultural indicators than by economic ones; H4 predicted that within individuals in the BCS70 cohort, the effect of education on completed fertility is mediated primarily by attitudinal pathways rather than by economic pathways.

The dissertation produced empirical verdicts on each hypothesis, three of which depart in substantively interesting ways from the predictions made at the outset. H1 was supported: the high-versus-low contrast in log eTFR is -0.187 ($p = 0.002$) across the 21-country panel, robust to four of five sensitivity specifications. H2 was supported parametrically and distributionally, with the U-shape emerging as the dominant gradient type across 12 of 21 countries and the minimum of the quadratic curve falling between the medium and high tiers. H3 was unsupported in the strong sense: at $N = 21$, neither the cultural nor the economic block produces a definitive ranking, and the LOOCV R^2 of -0.086 indicates that macro indicators lack reliable out-of-sample predictive power at this sample size. H4 was contradicted: for women in the BCS70 cohort, the economic pathway through earnings

accounts for 65.6% of the total causal effect of education on completed fertility, while the attitudinal pathway accounts for only 3.8%.

8.2 Summary of Contributions

The dissertation makes three distinct contributions to the literature on the demographic transition fitness paradox. The first is the systematic cross-country documentation of the U-shape as the dominant gradient type in contemporary Europe. The Stulp and Barrett (2016) review identified the J-curve hypothesis as theoretically important but empirically fragmentary; the gradient-shape classification in Chapter 4 and the formal quadratic and quantile analyses in Chapter 5 fill that gap on education-stratified period fertility data for 21 European countries over 2007–2024. The finding that 12 of 21 countries display the U-shape, with medium-educated women at the bottom of the fertility distribution, is a substantive refinement of the classical demographic-transition framing in which fertility falls monotonically with socioeconomic resources.

The second contribution is the application of Double Machine Learning causal mediation analysis to the BCS70 cohort. The combination of random-forest nuisance estimation, sample-splitting cross-fitting, and sequential mediator entry produces a transparent decomposition of the education–fertility relationship into attitudinal, economic, and partnership pathways. The finding that the economic pathway dominates by an order of magnitude — and the corresponding falsification of H4's attitudinal prediction — is a substantive constraint on the cultural-transmission account of fertility decline at the individual level. The methodological extension of DML mediation to BCS70 is, to my knowledge, the first application of this framework to this cohort's age-30 to age-42 sweeps.

The third contribution is the macro-level discovery–inference pipeline for H3 and the honest treatment of the disagreement between its two stages. Rather than reporting only the result that favoured the dissertation's preferred framework, Chapter 6 documents the Stage 1

SHAP ranking (which favours cultural indicators in aggregate, driven by freedom of religion) alongside the Stage 2 Bayesian ranking (which favours the economic model on WAIC, driven by GDP per capita), and treats the disagreement as the principal substantive finding. The LOOCV R^2 of -0.086 documents the limit of macro-level inference at $N = 21$ and constrains the strength of any cross-country causal claim the dissertation can support. The combination of methodological transparency and substantive honesty about the limits of $N = 21$ inference is itself a contribution to a literature in which cross-country regressions on small country samples are too often reported with more confidence than the data warrant.

8.3 Theoretical Implications

Three theoretical conclusions follow. The first is that the contemporary education–fertility gradient is non-monotonic at the population level: highly educated women do not have the lowest fertility, and the bottom of the distribution is located at medium education. Under the quantity–quality framework, this pattern is consistent with the prediction that medium-educated women face the steepest opportunity costs because they have meaningful wage penalties from childbearing but insufficient earnings to neutralise them through outsourced childcare, flexible work arrangements, or other fertility-supporting strategies that highly educated, high-earning women can afford. The dissertation's evidence does not establish this mechanism directly at the individual level, but the H4 finding that earnings dominate mediation is consistent with it.

The second theoretical implication is that the cultural-transmission framework, in its strong individual-level reading, receives less empirical support than the literature has assumed. The H4 finding that attitudinal pathways account for less than 4% of the education effect contradicts the most direct prediction the framework makes about within-individual mediation. A weaker reading of cultural-transmission — that low-fertility norms propagate through populations across cohorts via mechanisms that operate at population scales rather

than at the individual proximate-mechanism scale — remains tenable. The dissertation's data do not directly speak to the population-propagation question, but they do constrain the individual-mediation question and shift the burden of explanation toward economic and opportunity-cost mechanisms at the individual level.

The third theoretical implication concerns the demographic transition fitness paradox itself. Vining (1986) framed the paradox as a uniform negative gradient that required evolutionary explanation. Four decades of subsequent work, including this dissertation, have refined the empirical phenomenon to be more limited: the gradient is U-shaped rather than monotonic, the most highly educated women do not have the lowest fertility, the dominant individual-level mechanism is realised opportunity cost rather than evolved psychology systematically misfiring, and the cross-country variation in the gradient cannot be cleanly attributed to either cultural or economic macro indicators at currently feasible sample sizes. The paradox is therefore more limited than originally framed and more economically rational at the proximate level than the maladaptation framework suggested. Whether the paradox remains a paradox in the strongest sense — whether the rational opportunity-cost responses that depress contemporary fertility are nonetheless evolutionarily maladaptive over multigenerational fitness — is a question that the dissertation's single-generation evidence cannot resolve. The Goodman, Koupil and Lawson (2012) multigenerational finding suggests that the answer is yes; this dissertation does not directly speak to it but is consistent with it.

8.4 Final Remarks

The dissertation set out to test four hypotheses about the shape and mechanisms of the contemporary education–fertility gradient. It produced sharp empirical verdicts on all four, with two of those verdicts (H3 unsupported, H4 contradicted) departing substantively from the directional predictions made at the proposal stage. The willingness to state these verdicts cleanly, rather than to soften them in the direction of the predicted findings, is itself a feature

of the dissertation's contribution. Empirical demographic work too often selects specifications and reports findings that confirm the framework the analyst started with; this dissertation reports the findings the data produced, and treats the divergence between prediction and evidence as the principal occasion for theoretical refinement.

The broader debate over the demographic transition fitness paradox will not be resolved by any single dissertation. The empirical phenomenon is heterogeneous across populations, sub-groups, and historical periods; the theoretical frameworks competing to explain it (cultural transmission, quantity–quality, maladaptation) make overlapping predictions that are difficult to discriminate decisively in any one study. What the dissertation offers is a more rigorous and theoretically integrated test of the four most empirically tractable hypotheses than the existing literature provided, conducted with methods appropriate to the data structure at each level of analysis. The U-shape contribution at the population level, the DML mediation falsification at the individual level, and the honest reporting of the inferential limits at $N = 21$ in the macro analysis together constrain which theoretical accounts of the paradox remain viable and which require refinement. That is the modest but durable contribution this dissertation makes to a debate that has been open for forty years and will, in all likelihood, remain open for some time yet.